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8.3 Area of 2D Figures

Every area formula for a polygon can be built from the rectangle. A triangle is exactly half of the parallelogram formed by duplicating it and rotating the copy $180°$, so $A = \frac{1}{2}bh$. A trapezoid can be doubled the same way into a parallelogram with base $b_1 + b_2$, giving $A = \frac{1}{2}(b_1 + b_2)h$. A rhombus or kite splits along its diagonals into four right triangles that reassemble into a rectangle with sides d_1 and d_2, cut in half: $A = \frac{1}{2}d_1 d_2$. A regular polygon splits into n congruent triangles from the center, each with height equal to the apothem, so its total area is $A = \frac{1}{2} \cdot \text{apothem} \cdot \text{perimeter}$. This worksheet also uses composite-figure decomposition and **Heron's formula**, $A = \sqrt{s(s-a)(s-b)(s-c)}$ with $s = \frac{a+b+c}{2}$, for triangles where only the three side lengths are known.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Base 14, height 9. Area?

2. Base 11, height 6. Area?

3. Bases 7 and 13, height 4. Area?

4. Diagonals 9 and 12. Area?

5. Sides 8 and 5. Area?

6. Side 12. Area?

7. Base 20, height 7. Area?

8. Diagonals 6 and 10. Area?

9. Bases 10 and 18, height 6. Area?

10. Apothem 4, perimeter 30. Area?

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11. Base 15, height 8. Area?

12. Regular hexagon, side 2. Area?

13. Base 9, height 12. Area?

14. Use Heron's formula to find the area of a triangle with sides 9, 10, 17.

15. An L-shaped patio is a 12×9 rectangle with a 5×3 rectangular corner removed. Find the remaining area.

✓ Answer Key

1. 63
2. 66
3. 40
4. 54
5. 40
6. 144
7. 70
8. 30
9. 84
10. 60
11. 120
12. $6\sqrt{3}$
13. 54
14. 36
15. 93 square units